

Reuben Lavin

Incoming UBC Science student (Sept 2026), planning the transfer into Engineering. Robotics, embedded systems, and software.

reubenlavin08@gmail.com · reubenlavin@outlook.com · 604-773-1250 · Vancouver, BC
· reubenlavin08.github.io/portfolio · github.com/reubenlavin08 · linkedin.com/in/reuben-lavin



Education

University of British Columbia · Vancouver, BC Bachelor of Science · Engineering transfer planned after first year	Sept 2026 (incoming)
West Point Grey Academy · Vancouver, BC Grade 12 · 97% average · Honor Roll all years	2020–June 2026
• AP coursework: Computer Science A · Calculus BC · Statistics · English Language · Human Geography	

Selected Projects

full portfolio: reubenlavin08.github.io/portfolio

Assistive Navigation Helmet Wearable that maps obstacles to directional haptic feedback. Dual VL53L8CX time-of-flight sensors on independent I ² C buses and a BNO085 IMU at 100 Hz feed ESP32 firmware that converts an 8×8 depth grid into a three-motor haptic array with a distance-weighted urgency curve; verified on hardware with live desktop and phone point-cloud visualizers.	Robotics · Embedded
Spindle Whorl AR Mixed-reality preservation of a Coast Salish spindle whorl: photo-textured discs presented in a Unity 6 passthrough-AR gallery scene, built end to end and verified on Meta Quest hardware through a school device program.	Unity · Meta Quest 3

Experience & Leadership

Head of Technical Maintenance · Windsor Market On-call technician for a small commercial grocer	2021–2024
• Repaired industrial refrigeration, HVAC, electrical circuitry, and structural systems; also ran till, stock, and back-office operations across three years	
Black Belt & Assistant Instructor · Ken Shin Dojo / Kerrisdale CC Youngest black belt and assistant instructor in the dojo's history · kickboxing since 2021	2025–present
• Lead 20+ students across three weekly kickboxing and jiu-jitsu classes, including students with physical and mental disabilities	
Founder & Rhythm-Section Leader · WPGA Jazz Combo / R&B Band	2023–2026
• Founded the WPGA Jazz Combo (2023); grew from 4 to 7 members across 10+ performances • Co-founded and led the rhythm section of the WPGA R&B band; performed for Special Olympics Canada	
Grad Committee Treasurer · WPGA Class of 2026	2025–2026
• Managed the budget, vendor payments, and fundraising bookkeeping for the graduating class	

Awards & Certifications

Awards • UBC Physics Olympics · Bronze Medal, lab event (2026) • WPGA Citizenship Award • Alfred Lu Spirit Award • WPGA Merit Award (×2) • WPGA Music Program Hall of Fame • AP Scholars Award • Kumon Math · Outstanding Academics (2018–2023)	Certifications • Google AI Professional Certificate • Core Connectors Mental Health Literacy Certificate · Adam's Apples Foundation • CPRC First Aid Certificate • MMA Black Belt · Jiu Jitsu & Kickboxing • BC Novice Driver's Licence
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Activities & Skills

Activities • Varsity Tennis · WPGA · Vancouver Lawn Tennis & Badminton Club • Euclid Mathematics Contest 2026 • Volunteer · Kerrisdale Community Centre, Camp Jubilee restoration, WPGA productions	Software • Python · Java · SQL / SQLite • OpenCV · YOLOv8 · OpenVINO • Streamlit · Pandas • Git & GitHub · VS Code	Hardware • ESP32 · Arduino · ESP-IDF • Soldering · breadboarding · I²C wiring • RC platforms: servos, ESCs, foam-board airframes
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